Neural Networks 206 (2027) 109845

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

FusionAVP: A cross-attention- based heterogeneous feature fusion and contrastive learning strategy for antiviral peptide identification

Aoyun Geng [a], Jinwei Wang [a], Chunyan Cui [a], Yuanhao Li [a], Junlin Xu [b], Yajie Meng [c], Leyi Wei [d], Quan Zou [e], Zilong Zhang [a,*], Tao Wang [f,*], Feifei Cui [a,*]

[a] School of Computer Science and Technology, Hainan University, Haikou, 570228, China
[b] School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan, 430081, Hubei, China
[c] School of Computer Science and Artificial Intelligence, Wuhan Textile University, Wuhan, 430200, Hubei, China
[d] Centre for Artificial Intelligence driven Drug Discovery, Faculty of Applied Science, Macao Polytechnic University, Macao SAR, 999078, China
[e] Institute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu, 610054, China
[f] Laboratory of Protein Structure and Function, Institute of Medicine and Pharmacy, Qiqihar Medical University, Qiqihar, Heilongjiang, 161006, China

ARTICLE INFO

Keywords:
Heterogeneous feature fusion strategy
Contrastive learning
Cross-attention
Protein large language models
Antiviral peptides

ABSTRACT

Antiviral peptides (AVPs) are promising therapeutic candidates with broad-spectrum antiviral activity, low toxicity, and potential applications in antiviral drug discovery. Machine learning-based prediction of antiviral peptides can significantly reduce the workload in wet-lab experiments. However, current bio-sequence prediction approaches lack strategies to integrate protein large language models with traditional feature-based methods. Drawing from research in heterogeneous feature fusion and its application in image processing, we propose **FusionAVP**, a heterogeneous feature fusion strategy integrating contrastive learning for identifying antiviral peptides. To the best of our knowledge, FusionAVP is the first AVP prediction framework to introduce a cross-attention-based heterogeneous feature fusion module designed to capture the potential associations and complementarities among different feature encoding methods. It integrates protein large language models with contrastive learning to predict antiviral peptides. With fusion-based encoding, FusionAVP outperforms existing state-of-the-art models on public datasets. We further evaluate the differences among various large protein models and demonstrate the effectiveness of the proposed fusion strategy through visualizations. Finally, we conduct a case study to assess the model's interpretability and provide free access to a user-friendly web server at http://www.bioai-lab.com/FusionAVP.

1. Introduction

Viruses are microscopic pathogens that depend on host cells for reproduction and survival (Breitbart and Rohwer, 2005; Javanian et al., 2021). While not living organisms, they significantly affect host physiology during infection. Although hundreds of antiviral drugs have been developed, progress in pathogen research remains limited, and drug development is slow and often accompanied by side effects. As an alternative, therapeutic peptides—especially antiviral peptides (AVPs)—offer unique advantages. AVPs are short amino acid sequences with diverse structures and flexible conformations that effectively inhibit viral replication (Vilas Boas et al., 2019). Their mechanisms include disrupting viral membranes, blocking virus-host interactions, and suppressing genome replication. Compared to conventional drugs, AVPs show broad-spectrum antiviral activity, low toxicity, and fewer side effects (Skalickova et al., 2015). These properties make AVPs particularly suitable for vulnerable populations, such as the elderly and chronically ill. As research advances, AVPs are emerging as essential tools against viral infections, including drug-resistant strains (Qureshi et al., 2014).

With the development of bioinformatics and genomics, particularly advancements in sequencing technologies and machine learning, AI-based peptide identification methods have gained widespread application, providing more efficient and accurate tools for AVP discovery. Thakur et al. (Thakur et al., 2012) pioneered AVP prediction using a support vector machine (SVM) model that integrates peptide sequence features and physicochemical properties. Chang et al. (Chang and Yang, 2013) demonstrated the superiority of random forests in distinguishing

* Corresponding authors.
E-mail addresses: zhangzilong@hainanu.edu.cn (Z. Zhang), wangtao@qmu.edu.cn (T. Wang), feifeicui@hainanu.edu.cn (F. Cui).

https://doi.org/10.1016/j.neunet.2026.109845
Received 29 September 2025; Received in revised form 6 September 2026; Accepted 31 October 2026
Available online 31 October 2026
0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

AVPs using secondary structure features. To simplify prediction processes, AntiVPP 1.0 (Lissabet et al., 2019) developed a regression-based model capable of assessing AVPs' physicochemical properties. Similarly, Wei et al. (Wei et al., 2019) contributed to identifying potential functional peptide candidates through therapeutic peptide studies. Akbar et al. (Akbar et al., 2022) employed SHAP-based feature selection combined with a genetic algorithm-driven ensemble learning model to screen and validate AVPs. Schaduangrat et al. (Schaduangrat et al., 2019) trained six distinct machine learning algorithms to develop Meta-iAVP, a meta-predictor leveraging effective feature representation. Pang et al. (Pang et al., 2021) proposed a two-stage classification scheme specifically for predicting AVP activity against diverse viral families or species, incorporating Shapley-value-based interpretable machine learning to analyze descriptors' impact on antiviral activity. Incorporating deep learning, AI4AVP (Lin et al., 2022) used generative adversarial networks to augment positive samples in training datasets, enabling deep convolutional neural networks to learn from negative samples. Sharma et al. (Sharma et al., 2021) developed Deep-AVPpred, utilizing one-dimensional convolutional networks with multi-scale convolutional kernels and transfer learning of pre-trained embeddings. Kurata et al. (Kurata et al., 2022) introduced iACVP, a computational tool leveraging W2V dictionaries and optimized amino acid k-mer sizes for enhanced prediction. The iACVP-MR (Liang et al., 2025;) model combined multiple features with long short-term memory (LSTM) networks and bidirectional gated recurrent units (BiGRU) for feature engineering. FFMAVP (Cao et al., 2023) extracted amino acid and sequence descriptors through a two-stage task module for fusion and prediction. AutoPeptideML (Fernandez-Diaz et al., 2023) provided a web server for users to build custom models based on peptide bioactivity while offering prediction reliability insights. Guan et al. (Guan et al., 2024) utilized contrastive learning and multi-feature strategies to enhance model interpretability and predictive capability. Most recently, DeepAVP-TPPRed (Ullah et al., 2024) employed image-based feature extraction and binary tree growth algorithms to overcome machine learning feature engineering limitations. Lastly, Deepstacked-AVPs (Akbar et al., 2024) integrated novel position-specific scoring matrices, word2vec-based semantic features, and physicochemical properties through information gain and stacked ensemble learning, achieving significant improvements in AVP classification performance.

Despite these advancements, existing AVP identification methods still face challenges in prediction accuracy. Most approaches have yet to fully explore the application of large language models (LLMs) (Luo et al., 2025; Xu et al., 2024). However, in other biological sequence prediction tasks (Geng et al., 2025; Xiao et al., 2024; Zhu et al., 2023), studies have demonstrated that LLMs significantly improve prediction accuracy and reliability. By learning from vast biological sequence data, LLMs capture complex patterns and potential biological functions, offering highly precise prediction capabilities for peptides (Perez et al., 2023). Their semantic understanding has proven effective in both protein (Du et al., 2023) and RNA sequences, suggesting that incorporating LLMs into AVP identification could substantially enhance accuracy and drive antiviral drug development further. Attention mechanisms have also been widely applied in biological sequence classification tasks. Compared with traditional feature extraction methods, attention-based models can dynamically assign different weights to residues or sequence regions, allowing them to capture informative local patterns and long-range dependencies within protein or peptide sequences. In particular, Transformer-based protein language models use self-attention to learn contextual representations from large-scale protein sequences, which has improved performance in many downstream prediction tasks.

Moreover, existing prediction methods often combine LLM-extracted features with traditional features, usually by simply concatenating them (Du et al., 2024; Jia et al., 2024; Li et al., 2024). However, this approach does not fully account for the different semantic information contained in these features, potentially overlooking their associations and complementarities. Traditional features focus on explicit local biological information (Yan et al., 2024), while LLM features capture complex global patterns. Simple concatenation may result in information loss or conflict, affecting prediction accuracy. Similar challenges have been observed in applications such as drug-target interaction (Wang et al., 2022) and affinity prediction (Wu et al., 2024), prompting widespread adoption of cross-attention (Li et al., 2024) mechanisms for multimodal embedding fusion. These mechanisms effectively integrate features from diverse sources, improving model performance.

Inspired by this, we developed **FusionAVP**, an algorithm designed to predict AVPs by more precisely integrating traditional features with representations from LLMs. The core component is a cross-attention-driven heterogeneous feature fusion module, in which a contrastive learning strategy is incorporated to enhance feature alignment and discrimination. Our algorithm outperformed existing AVP prediction algorithms across various metrics. Additionally, through comparative and validation experiments, we identified optimal feature combinations to achieve better predictions. Ablation experiments verified the fusion module's effectiveness, highlighting its importance in improving prediction performance (Fig. 1E). Finally, a case study validated the model's efficacy and practicality in real-world applications. To facilitate use by researchers, we developed a web server for FusionAVP, accessible via http://www.bioai-lab.com/FusionAVP. This platform allows researchers to easily utilize the model for AVP predictions.

2. Materials and methods

2.1. Overview of FusionAVP

As shown in Fig. 1, this study aims to enhance the representation of antiviral peptide sequences by integrating two types of features: general features from large pre-trained models and traditional features based on biological knowledge. The former captures global structural and semantic patterns, while the latter encodes domain-specific attributes. To combine these complementary features, we employed a cross-attention-driven heterogeneous feature fusion strategy, generating unified embeddings that reflect both global context and biological priors. Next, we used the two sets of embeddings derived from the large model-based features as positive sample pairs in contrastive learning. The contrastive learning process optimized the embeddings by comparing the differences between positive and negative sample pairs, with all other samples in the batch serving as negative samples. This process enhanced the model's ability to distinguish between AVPs and non-AVPs. Subsequently, we trained and fine-tuned the model using both contrastive loss and binary cross-entropy (BCE) loss, which jointly optimized representation learning and classification performance, resulting in more accurate antiviral peptide predictions.

2.2. Dataset

For the dataset, in order to better validate the model's ability to detect antiviral activity, we selected the largest available dataset. We utilized the antiviral peptide dataset collected by Pinacho-Castellanos et al. (Pinacho-Castellanos et al., 2021), and used the starPep tool (Aguilera-Mendoza et al., 2023) to gather a series of antimicrobial peptides from starPepDB. From this collection, we selected peptides with antiviral activity, retaining those with a length between 5 and 100 natural amino acids, and removed peptides with multiple antimicrobial activities. In the end, 2944 antiviral peptide sequences were obtained for training and testing. Negative samples were sourced from the UniProt (Consortium, 2015) database, with sequences filtered by CD-HIT (Fu et al., 2012) to remove those with >50% sequence identity, ensuring the independence and diversity of the dataset. The division of training and test sets is detailed in Table 1 (also accessible at http://www.bioai-lab.com/FusionAVP).

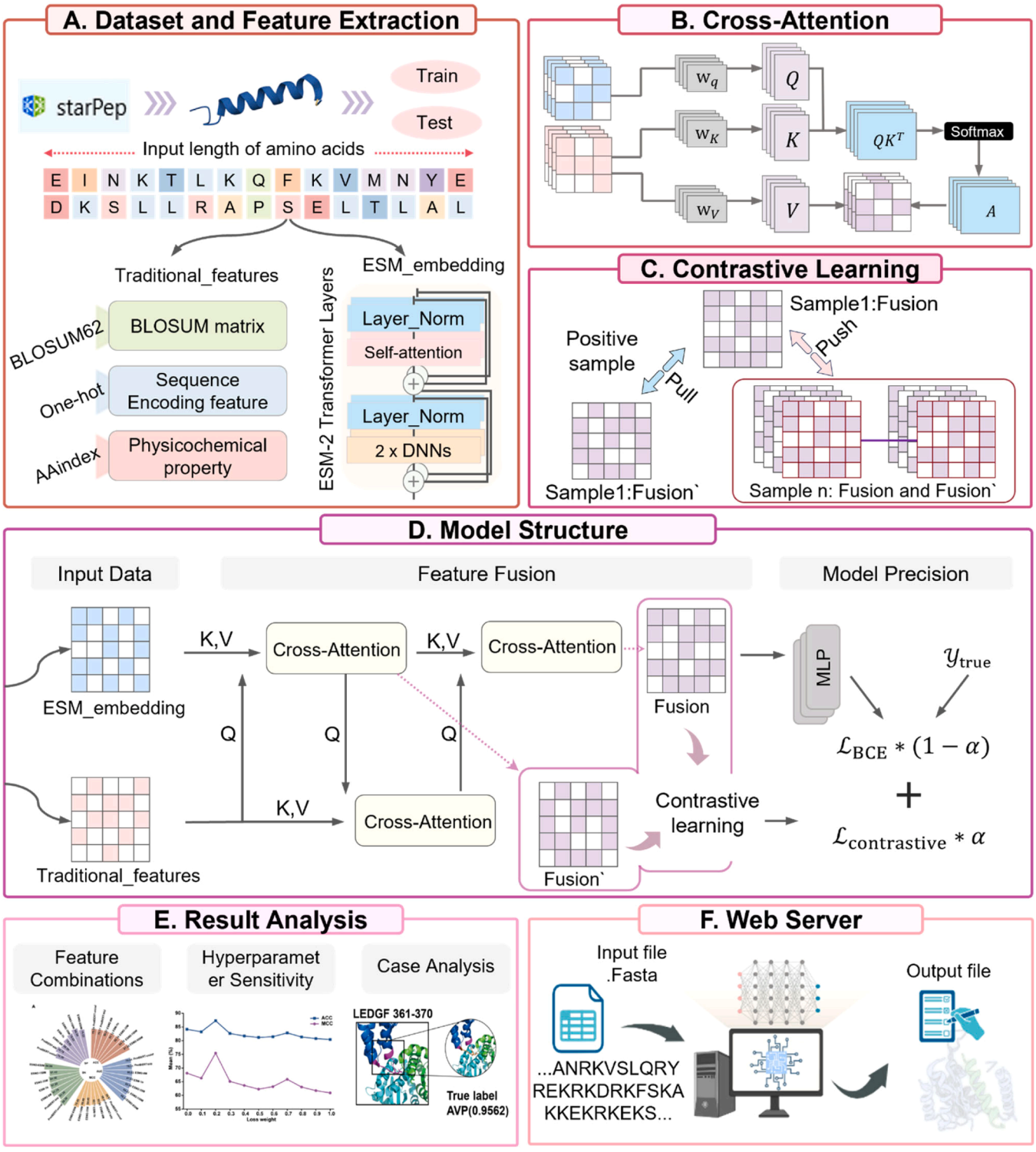


Fig. 1. Overview of the proposed FusionAVP framework. **A.** Dataset and Feature Extraction: Antiviral peptide data is collected from starPep, filtered, and split into training and test sets. Traditional features and deep representations from ESM-2 are extracted for each sequence. B. Cross-Attention: The computational process of the Cross-Attention mechanism. **C.** Contrastive Learning: Fusion and Fusion′ of the same sample are treated as positive pairs and pulled closer, while other views in the batch serve as negatives and are pushed apart. **D.** Model Structure: Input features are processed by the proposed feature fusion module and optimized using a weighted combination of BCE loss and contrastive loss. **E.** Result Analysis: Includes feature combination selection, hyperparameter sensitivity, and case studies. **F.** Web Server: Users can upload FASTA-formatted sequences via the web interface; the system outputs predictions indicating whether each peptide is antiviral.

2.3. Feature extraction

In our approach, we treat traditional features and general features from large pre-trained models as two heterogeneous feature views of the same amino acid sequence, as shown in Fig. 1A. Traditional features offer greater interpretability, allowing clear understanding of the origin and meaning of each dimension. However, they lack the ability to capture deep semantic patterns and global structural information—advantages provided by large pre-trained models. Since the peptide sequences used in this study ranged from 5 to 100 amino

Table 1
The division of training and test sets for the antiviral peptides we used.

	Dataset	Positive	Negative	Size
Antiviral	training	2321	2321	4642
	test	623	623	1246

acids, we set the maximum sequence length to 100 for all residue-level feature matrices. For traditional features, each sequence was encoded as a residue-level matrix and zero-padded to length 100. For ESM-2 embeddings, residue-level representations were extracted after removing special tokens and aligned with traditional features according to the original amino acid positions. No truncation was required because sequences longer than 100 amino acids were excluded during preprocessing.

2.3.1. Traditional feature extraction

Based on the methods used for traditional feature extraction, we categorize the features into three types: BLOSUM matrix-based features, Sequence Encoding Features, and Physicochemical Property-based Features.

BLOSUM Matrix-Based Features: The BLOSUM matrix (Fang et al., 2023; Tung et al., 2007) captures evolutionary information to analyze conserved regions and relationships in amino acid sequences, helping to distinguish peptides with antiviral activity. Originally introduced by Henikoff et al. (Henikoff and Henikoff, 1992), the BLOSUM matrix computes substitution scores between amino acid pairs based on aligned homologous protein sequences, reflecting the likelihood of evolutionary substitutions. Each value in BLOSUM62 represents the substitution score between two of the 20 standard amino acids, with higher scores indicating higher substitution likelihood. The number in BLOSUM matrices (e.g., BLOSUM50, BLOSUM62, BLOSUM80) denotes the sequence similarity threshold used in their construction. In this study, we apply BLOSUM62 to generate an $i \times 20$ feature matrix for each sequence, where i is the sequence length.

$$BLOSUM62 = \begin{bmatrix} V_{1,1} & \cdots & V_{1,20} \\ \vdots & \ddots & \vdots \\ V_{20,1} & \cdots & V_{20,20} \end{bmatrix} \quad (1)$$

Sequence Encoding Features: Sequence encoding features are derived using binary encoding (One-hot encoding) to represent amino acid sequences. This method transforms categorical data into numerical vectors and is widely used in biological prediction tasks due to its strong generalizability (Wang et al., 2023). In one-hot encoding, each amino acid is represented by a 20-dimensional binary vector with a single 1 and all other entries set to 0, effectively capturing the positional and frequency patterns of residues in a sequence. For the 20 standard amino acids:

- The first amino acid "A" is represented as: [1, 0, 0, …, 0],
- The second amino acid "C" is represented as: [0, 1, 0, …, 0],
- ……,
- The last amino acid "Y" is represented as: [0, 0, 0, …, 1].

Physicochemical Property-Based Features: The AAindex database (Kawashima et al., 2007) offers comprehensive data on the physicochemical and biological properties of amino acids, serving as a key resource for analyzing protein structure and function. In this study, we selected 12 amino acid properties related to antiviral activity from AAindex to characterize AVPs from multiple perspectives. To assess viral membrane binding capacity, we included net charge (KLEP840101), accessible surface area (BIOV880101), transfer energy (NOZY710101), and amino acid composition in membrane and transmembrane proteins (NAKH900109, NAKH920108). For evaluating peptide stability and function, we used alpha-helix and beta-strand propensities (MAXF760101, BLAM930101, LIFS790101), relative partition energy (MIYS990104), and composition in intracellular proteins (CEDJ970104). To reflect adaptability in complex environments, we selected hydrophilicity/hydrophobicity balance (TSAJ990101) and entropy change during structure formation (HUTJ700103). These features were encoded into an $i \times 12$-dimensional matrix, where i is the peptide length.

2.3.2. Large model-based feature extraction

For large language models (Han et al., 2022), we used the pre-trained Transformer-based model ESM-2 (Lin et al., 2023), developed by the Facebook AI research team, which is specifically designed for protein sequence analysis. The model is trained on protein sequences from the UniRef (Suzek et al., 2007) database and uses the Self-Attention mechanism (as shown in the formula below) to adjust the weights of contextual relationships between amino acids in a sequence, thereby effectively learning the dependencies between amino acid residues (Fu et al., 2024).

$$Self - Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

Here, Q is the query matrix, K is the key matrix, V is the value matrix, and d_k is the dimension of the keys, used for normalization. In the multi-head self-attention mechanism, Q, K, and V are matrices derived from the same input data, mapped through learnable parameters in the fully connected layer.

Furthermore, ESM-2 is trained on large-scale evolutionary data, enabling it to capture conserved features and mutation patterns by comparing homologous protein sequences. During training, 85% of amino acids are used to predict the masked 15%, allowing the model to learn deep, evolution-based representations that enhance accuracy and robustness in peptide prediction. ESM-2 is available in multiple parameter sizes—8 M, 35 M, 150 M, 650 M, 3B, and 15B—corresponding to different numbers of Transformer layers. While larger models capture more complex features, they require more computational resources and are more susceptible to overfitting. In this study, we selected models up to 650 M parameters to balance performance and efficiency for AVP prediction.

2.4. Heterogeneous feature fusion strategy

Different feature views of the same amino acid sequence can encode distinct and complementary information. Although both traditional features and those from large pre-trained models are based on sequential data, they differ significantly in semantic depth. Antiviral peptide sequences are usually short, and their activity identification depends on the joint contribution of multi-level sequence information. ESM-2 embeddings provide contextual semantic representations, latent evolutionary patterns, and global sequence dependencies learned from large-scale protein sequences, whereas traditional features such as BLOSUM62, one-hot encoding, and AAindex encode biologically interpretable local information, including amino acid substitution propensity, residue composition and positional information, and physicochemical properties. Therefore, these two types of representations are complementary but heterogeneous. Direct concatenation may be insufficient to model their interactions and may introduce information redundancy or semantic interference.

To address this issue, we propose a cross-attention-based fusion module that effectively captures and integrates complementary information between the two features. The Cross-Attention mechanism is widely used in multimodal learning (Huang et al., 2019). Unlike the Self-Attention mechanism, Cross-attention models interactions between different feature representations. It computes attention weights between two feature views using learnable parameters, with the query (Q) from one feature view and the key (K) and value (V) from the other (as depicted in Fig. 1B). In this way, the cross-attention mechanism

facilitates effective information exchange between heterogeneous features. The computation process is as follows:

$$Cross-Attention(Q,K,V) = softmax\left(\frac{Q^1 K^{2^T}}{\sqrt{d_{k^2}}}\right)V^2 \quad (3)$$

where Q is from one view, and K and V are from another.

In the proposed feature fusion module (Fig. 1D), we initially assign LLM features as **K** and **V**, and traditional features as **Q**, using the latter to refine the former. A three-stage heterogeneous feature fusion mechanism with iterative role exchange is employed to progressively strengthen the interaction between global semantic representations and biologically grounded local features, thereby facilitating the integration of heterogeneous feature and improving model performance. In the first stage, traditional features are used as the query, while ESM-2 embeddings serve as the key and value, allowing biological prior information to guide the extraction of AVP-relevant semantic patterns from the large model representations. In the second stage, the interaction direction is reversed, so that contextual information from the large model representations can be fed back into the traditional feature space and enhance its expressive capacity. In the third stage, the two refined representations are further integrated to generate a joint embedding that contains both global semantic information and biological prior knowledge. Compared with one-step concatenation, this progressive interaction strategy can establish more fine-grained associations between different levels of sequence information and gradually enhance discriminative features related to antiviral activity. The resulting fused embeddings are then used for AVP prediction and classification. Since features extracted by the large language model contain deeper semantic information, we treat the two outputs—F and F`—produced by the heterogeneous feature fusion strategy (with ESM-2 embeddings as value vectors **V**) as positive sample pairs in contrastive learning (Yin et al., 2023), as illustrated in Algorithm 1. This further enhances the model's robustness and predictive performance. Specifically, we construct positive and negative sample pairs for training using the Fusion and Fusion' outputs provided by the feature fusion module. We treat the two embeddings of the same sample as a positive sample pair and the embeddings of the other samples in the same batch as negative samples (Fig. 1C). Although F and F′ are derived from the same peptide sequence, they are generated through different heterogeneous feature interaction paths and therefore capture different levels of information. Specifically, F′ is obtained by using traditional features as the query and ESM-2 embeddings as the key and value, which makes it a traditional -feature-guided semantic view of the large-model representation. In contrast, F is generated after the subsequent reverse interaction and final fusion stage, and therefore represents a bidirectionally refined joint view that integrates ESM-2 contextual semantics with biological priors from the traditional feature space. Thus, F and F′ share the same semantic identity of the input sample but emphasize different heterogeneous feature interaction patterns, making them suitable as complementary views for contrastive learning. In the latent space, we improve the discriminative ability of the embeddings by maximizing the similarity between positive sample pairs while minimizing the distance between negative sample pairs.

Algorithm 1
Cross-Attention-Driven Heterogeneous Feature Fusion Strategy.

Input: ESM-2 Embeddings **E**, Traditional features **M**
Output: Fused representation, $\boldsymbol{L}_{\mathbf{contrastive}}$
Step1: Compute cross-attention from Traditional features to ESM-2 Embeddings:
H_1 $(F`) \leftarrow$ *Cross-Attention* $(Q = M, K = E, V = E)$
Step2: Compute cross-attention from $\mathbf{H_1}$ to Traditional features:
$H_2 \leftarrow$ *Cross-Attention* $(Q = H_1, K = M, V = M)$
Step3: Compute cross-attention from $\mathbf{H_2}$ to $\mathbf{H_1}$:
$F \leftarrow$ *Cross-Attention* $(Q = H_2, K = H_1, V = H_1)$
Step4: Project **F** and H_1 into the same latent space.
$F_1, F_2 \leftarrow g(F), g(H_1)$ ▷ Eq.4
Step5: Compute contrastive loss:
$L_{contrastive\ loss} \leftarrow \sum_{i=1}^{N} \left[l(F_1{}^i, F_2{}^i) + l(F_2{}^i, F_1{}^i)\right]$ ▷ Eq.5
Return: **F**, $\boldsymbol{L}_{\mathbf{contrastive}}$

First, we input the outputs of Fusion (view F_1) and Fusion' (view F_2) into a shared-weight projection layer g. The role of the projection layer g is to map the embeddings from the two views into a common latent feature space, where the contrastive loss can be computed. The projection layer g is implemented by a two-layer MLP:

$$g(F) = w_2\ (w_1 F + b_1) + b_2 \quad (4)$$

where 1 and 2 denote different layers, w represents the learnable weight matrices, and b represents the bias terms. To maximize the consistency of the representations of the same sample across different views, we adopt a contrastive learning strategy, which is implemented as follows: the embedding of sample i in the F_1 view is used as the anchor, the embedding in the F_2 view as the positive sample, and the embeddings of the other samples in the same batch as negative samples. We use cosine similarity to measure the similarity between samples and calculate the contrastive loss using the following formula:

$$l(F_1{}^i, F_2{}^i) = -log\frac{e^{\frac{\theta(F_1{}^i, F_2{}^i)}{\tau}}}{e^{\frac{\theta(F_1{}^i, F_2{}^i)}{\tau}} + \sum_{j\neq i}\left(e^{\frac{\theta(F_1{}^i, F_1{}^j)}{\tau}} + e^{\frac{\theta(F_1{}^i, F_2{}^j)}{\tau}}\right)} \quad (5)$$

where τ is the temperature parameter, and $\theta(F_1{}^i, F_2{}^i) = \cos(g(F_1{}^i), g(F_2{}^i))$ represents the cosine similarity between the two embeddings. $(F_1{}^i, F_1{}^j)$ denotes negative sample pairs from the same view, while $(F_1{}^i, F_2{}^j)$ represents negative sample pairs from different views.

Since we sum the contrastive loss rather than average it, batch size significantly affects the results. As it determines the number of negative samples, batch size influences the balance between positive and negative learning signals, thereby impacting model performance. Minimizing the contrastive loss reduces the distance between positive pairs and increases it between negative pairs, leading to more robust embedding representations.

2.5. Model training strategy

We employed a two-layer Multi-Layer Perceptron (MLP) for classification and optimized the model using Binary Cross-Entropy (BCE) Loss, a standard objective for binary tasks. BCE measures the discrepancy between predicted probabilities and true labels, and is defined as follows:

$$L_{BCE\ Loss} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1-y_i)\log(1-p_i)] \quad (6)$$

where N is the number of samples, y_i is the true label of the i th sample (either 0 or 1), and p_i is the predicted probability that the i th sample belongs to class 1. To jointly optimize the feature representation learning and classification prediction tasks, we combined the contrastive loss and BCE Loss in a weighted manner. The final loss function is defined as:

$$L_{total} = \alpha \cdot L_{contrastive_{loss}} + (1-\alpha)\cdot L_{BCE\ Loss} \quad (7)$$

where α is a hyperparameter used to control the weight balance between the contrastive loss and BCE loss. By adjusting this parameter, we can balance their contributions during the optimization process, thereby improving the model's performance in both feature representation learning and classification prediction tasks. In this experiment, ReLU is used as the activation function, with a dropout rate of 0.5 applied to all hidden layers. The model is optimized using the Adam optimizer with a learning rate of 0.0001, batch size of 64, $\alpha = 0.2$, and trained for 70 epochs. All experiments were conducted on an NVIDIA A100 GPU.

The joint optimization of contrastive loss and BCE loss also helps

reduce the risk of representation collapse. In the contrastive learning module, representations from other samples in the same mini-batch are used as negative samples, which prevents the model from only maximizing the similarity between positive pairs and encourages separability among different samples. Meanwhile, the BCE loss provides explicit class-supervised signals for AVP/non-AVP classification, further constraining the representation space. Therefore, the model is encouraged to align two complementary views of the same sample while maintaining discriminative boundaries between AVP and non-AVP samples.

2.6. Evaluation metrics and methods

To evaluate the model's performance, we used five standard statistical metrics (Wang et al., 2024): Accuracy (ACC), Matthew's Correlation Coefficient (MCC), Specificity (SP), Sensitivity (SN), and the area under the Receiver Operating Characteristic curve (AUC). These metrics are defined as follows:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TN + FN) \times (TP + FP) \times (TN + FP) \times (TP + FN)}} \tag{9}$$

$$SN = \frac{TP}{TP + FN} \tag{10}$$

$$SP = \frac{TN}{TN + FP} \tag{11}$$

$$AUC = \int_0^1 ROC\ curve \tag{12}$$

where TP and TN represent correctly predicted positive and negative samples, FP and FN denote incorrect predictions. AUC evaluates the model's ability to distinguish between classes—values closer to 1 indicate excellent discrimination, whereas values near 0.5 suggest no discriminative power.

3. Results and discussion

3.1. Comparison with state-of-the-art methods

Many models have been developed for AVP recognition and prediction using various encoding methods and ensemble classifiers (Fernandez-Diaz et al., 2023; Liang et al., 2025;; Pang et al., 2021; Schaduangrat et al., 2019; Sharma et al., 2021; Ullah et al., 2024). However, these models typically combine all encodings together in a joint manner (Akbar et al., 2024; Nath, 2021; Thakur et al., 2012). In contrast, we propose using a feature fusion module and compare it with SOTA models. We compare our method with several state-of-the-art models on an independent test set. To evaluate the robustness of FusionAVP, we conducted repeated experiments using 10 different random seeds under the same training and testing settings, and reported the mean ± standard deviation of the 10 runs as the final performance of FusionAVP. In addition, we provide the precision values and the corresponding confusion matrix counts for each independent run in Supplementary Table S1. Across the runs, 28–35 false negatives and 124–132 false positives were observed, indicating that the high sensitivity, while enabling the identification of more positive samples, was also accompanied by a non-negligible number of false-positive predictions. As shown in Table 2, FusionAVP achieved small standard deviations across different evaluation metrics, indicating that the model is robust to random initialization and that the observed performance improvement was not caused by a single favorable run. The results show that the proposed FusionAVP achieved overall performance improvements compared with the other models. Compared with Uni-DL4BioPep, which also uses the ESM-2 large protein language model for feature extraction, FusionAVP improved ACC and MCC by 3.08% and 6.05%, respectively. In addition, it is worth noting that the dataset used in this study was entirely collected from publicly available sources and was not specifically filtered or optimized for any particular model, further demonstrating the stability and applicability of the proposed model.

3.2. Evaluating the effectiveness of feature fusion strategy

To evaluate the effectiveness of the proposed feature fusion module, we used UMAP (McInnes et al., 2018) to visualize the differences between simple concatenation and our fusion strategy. UMAP provides nonlinear dimensionality reduction while preserving both local and global structures, making it well-suited for analyzing the distribution of high-dimensional embeddings. As shown in Fig. 2, we compared the UMAP projections of traditional features concatenated with large model embeddings and those processed by the fusion module. In the concatenation case, yellow (positive) and blue (negative) samples are heavily overlapped, showing poor class separability. In contrast, the fusion module yields a clearer boundary with reduced overlap, indicating that it effectively captures and integrates complementary information from both views, improving inter-class discrimination. To provide quantitative support for the UMAP visualization, we further calculated the silhouette score using the class labels of AVP and non-AVP samples. The silhouette score increased from 0.16 before fusion to 0.45 after fusion, indicating improved inter-class separation in the fused representation space. This result is consistent with the UMAP visualization, where the fused features show clearer boundaries and reduced overlap between positive and negative samples.

To further evaluate the advantage of the proposed fusion strategy, we compared FusionAVP with two representative fusion methods, namely bilinear pooling and gated fusion, in addition to simple concatenation. As shown in Table 3, the proposed feature fusion strategy achieved the best overall performance, outperforming the other fusion methods in terms of ACC, MCC, SN, and AUC. Although its SP was slightly lower than that of simple concatenation, its overall discriminative ability was clearly stronger, indicating that the proposed fusion strategy can more effectively identify AVP samples. Furthermore, although bilinear pooling can model multiplicative interactions between heterogeneous features, it usually introduces more parameters and higher computational complexity, which may lead to overfitting or unstable performance on relatively limited peptide sequence datasets. Gated fusion can adaptively regulate the contributions of different features, but it mainly focuses on feature-weight allocation and has relatively limited ability to capture fine-grained residue-level interactions across views. These results demonstrate its superior ability to integrate heterogeneous features, its adaptability across various feature combinations, and its strong potential for practical applications in antiviral peptide prediction and beyond. In addition, to further investigate the feature fusion strategy and the contribution of different feature streams, we conducted an additional ablation analysis. Specifically, while keeping the full model architecture and cross-attention layers unchanged, one feature stream was replaced with zero vectors, allowing us to more clearly assess the individual contribution of each feature stream to model performance. As shown in Table 4, when the ESM-2 embeddings were replaced with zero vectors, the ACC further decreased to 81.18%. Meanwhile, the SP remained relatively high under this setting, indicating that the model retained a strong ability to identify negative samples when relying solely on handcrafted features. This result suggests that handcrafted features preserve certain discriminative information, but are insufficient on their own to achieve the performance of the full model. In contrast, when the handcrafted features were replaced with zero vectors, the ACC decreased from 87.28% to 84.62%, indicating that ESM-2 embeddings provide stronger semantic

Table 2
Comparison of the metrics of our proposed model with those of other models.

Method	ACC (%)	MCC (%)	SP (%)	SN (%)	AUC (%)
ABPDiscover (Pinacho-Castellanos et al., 2021)	82.8	66.2	89.2	76.4	89.6
ESM-1b and iLearn and ProtDCal (García-Jacas et al., 2022)	84.2	68.8	89.9	78.5	N/A
UniDL4BioPep (Du et al., 2023)	84.2	69.4	79	91.6	90.7
AutoPeptideML (Fernandez-Diaz et al., 2023)	80.12	55.65	83.3	86.67	85.06
FusionAVP(our)	87.28 ± 0.39	75.45 ± 0.77	79.62 ± 0.48	94.94 ± 0.35	88.41 ± 0.30

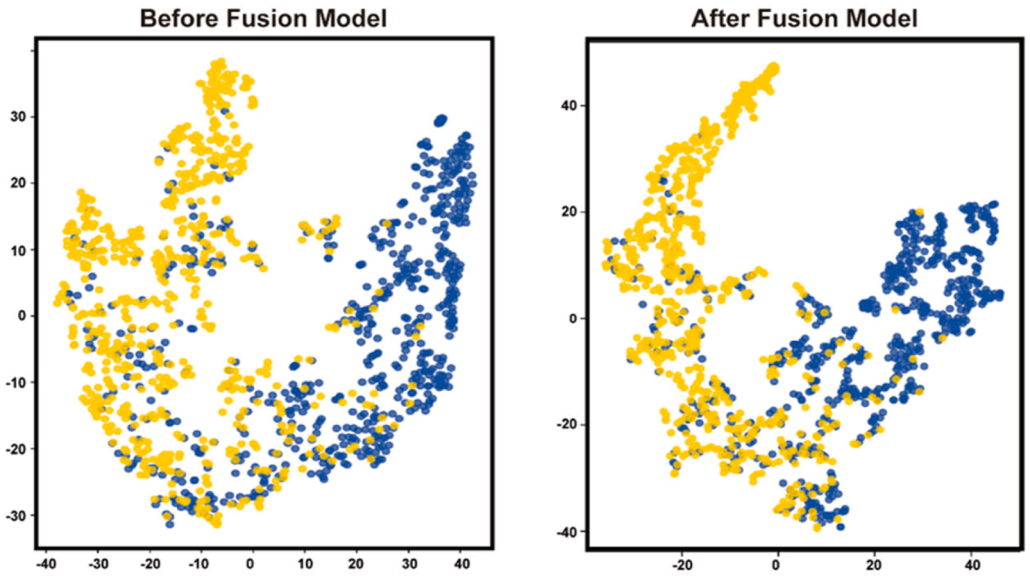


Fig. 2. Comparison between simple concatenation of traditional and large model features and the proposed fusion module. Yellow denotes positive samples; blue denotes negative samples.

Table 3
Performance comparison of different fusion strategies on the independent test set.

Fusion strategy	ACC(%)	MCC(%)	SP(%)	SN(%)	AUC(%)
Simple concatenation	83.06 ± 0.47	66.39 ± 0.92	80.98 ± 0.66	85.14 ± 0.53	83.22 ± 0.41
Bilinear pooling	84.28 ± 0.44	69.05 ± 0.86	79.72 ± 0.64	88.84 ± 0.55	85.08 ± 0.38
Gated fusion	85.82 ± 0.42	72.79 ± 0.80	80.56 ± 0.58	91.08 ± 0.50	86.89 ± 0.35
FusionAVP(our)	87.28 ± 0.39	75.45 ± 0.77	79.62 ± 0.48	94.94 ± 0.35	88.41 ± 0.30

representations than handcrafted features, while still benefiting from the incorporation of biologically interpretable handcrafted features.

3.3. *Effectiveness of various feature combinations in AVPs prediction*

To investigate the impact of combining protein LLM features with traditional encoding methods in AVP recognition, we integrated three traditional feature types—Sequence Encoding Feature based on sequence information, BLOSUM matrix encoding based on evolutionary information, and AAindex encoding based on physicochemical properties—with various pretrained protein LLMs. The selected LLMs include ProtBERT-bfd, ProtBERT-uniref (Elnaggar et al., 2021) (differing in the pretraining data), ESM-1b (Rives et al., 2021) and multiple versions of ESM-2 (Lin et al., 2023) (8 M, 35 M, 150 M, and 650 M parameters). Each traditional encoding method was paired with the seven LLMs to generate different feature combinations, which were subsequently evaluated on the independent test set using multiple performance metrics. The results are reported as the mean ± standard deviation across 10 different random seeds (Tables S2–S4).

As shown in Fig. 3A–C, ESM-2 (650 M) achieved the highest mean ACC, MCC, and AUC within each of the three traditional feature settings. Among the evaluated combinations, ESM-2 (650 M) paired with BLOSUM62 yielded the highest mean ACC of 87.28 ± 0.39%, MCC of 75.45 ± 0.77%, and AUC of 88.41 ± 0.30%. With ESM-2 (650 M) held fixed, BLOSUM62 exceeded one-hot encoding by 4.16 percentage points in ACC and 5.45 percentage points in AUC, whereas AAindex exceeded one-hot encoding by 0.46 and 1.16 percentage points, respectively. The sensitivity–specificity patterns also differed between feature types.

Table 4
Ablation analysis of feature-stream contributions on the independent test set.

Input configuration	ACC (%)	MCC (%)	SP (%)	SN (%)	AUC (%)
Traditional features replaced with zeros	84.62 ± 0.45	69.67 ± 0.82	79.10 ± 0.62	90.14 ± 0.51	86.31 ± 0.38
ESM-2 embeddings replaced with zeros	81.18 ± 0.51	62.36 ± 0.91	80.52 ± 0.68	81.84 ± 0.63	82.74 ± 0.44
FusionAVP(our)	87.28 ± 0.39	75.45 ± 0.77	79.62 ± 0.48	94.94 ± 0.35	88.41 ± 0.30

AAindex achieved higher sensitivity than one-hot encoding (90.24% versus 87.20%) but lower specificity (76.92% versus 79.04%). BLOSUM62, by contrast, achieved both higher sensitivity (94.94%) and higher specificity (79.62%) than one-hot encoding. These results identify ESM-2 (650 M) paired with BLOSUM62 as the strongest combination in terms of mean ACC, MCC, and AUC among those evaluated, while showing that the balance between sensitivity and specificity depended on the feature combination.

3.4. Impact analysis of model parameters

We examined the effects of batch size and the loss weight α on FusionAVP performance. Batch sizes of 8, 16, 32, 64, 128, and 256 were evaluated, while α was varied from 0 to 1 in increments of 0.1. Batch size determines the number of in-batch negative pairs available for contrastive learning, whereas α controls the relative contributions of the contrastive and classification losses. Because the contrastive loss was

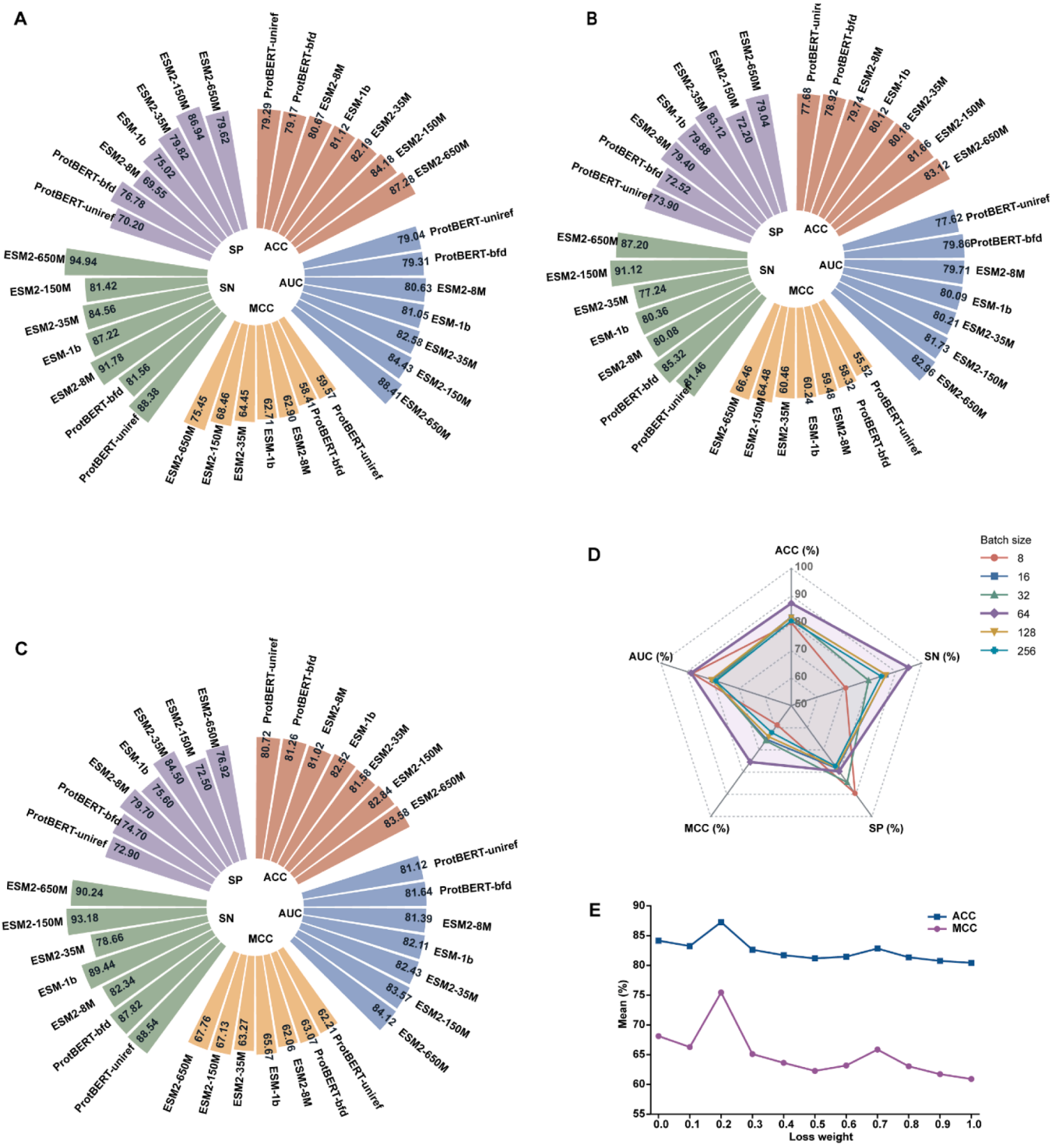


Fig. 3. Comparison of feature combinations and hyperparameter sensitivity of FusionAVP. A–C, Performance comparison of seven protein language models combined with BLOSUM62-based features (A), one-hot sequence encoding features (B), and AAindex-based features (C). D, Comparison of model performance under different training batch sizes. E, Effect of varying the loss function weight α on model performance.

summed rather than averaged, its magnitude also depended on batch size. Each setting was evaluated on the independent test set across 10 random seeds, with results reported in Tables S5 and S6.

As shown in Fig. 3D and Table S5, a batch size of 64 achieved the highest mean ACC, MCC, SN, and AUC. A batch size of 8 yielded the highest specificity but substantially lower sensitivity. Increasing the batch size beyond 64 did not improve performance, indicating that a larger pool of in-batch negatives did not necessarily yield better predictions. For the loss weight α, the highest mean ACC and MCC were obtained at $\alpha = 0.2$ (Fig. 3E and Table S6). Both metrics were lower at all other evaluated values, although the changes were not monotonic. These results show that performance was sensitive to both batch size and loss weighting, with the best observed settings being a batch size of 64 and $\alpha = 0.2$.

3.5. Interpretability through model performance and case analysis

To better evaluate the performance and interpretability of FusionAVP, we first analyzed the training and validation dynamics. As shown in Fig. 4A, the model's accuracy steadily increased while loss decreased over epochs, eventually stabilizing at epoch 70. This indicates effective optimization without overfitting. We then tested the model on an independent test set. Fig. 4B shows the probability density plots, where positive samples are concentrated near 1.0 and negative samples near 0.0, demonstrating strong discriminative capability. While a small overlapping region exists near the 0.5 threshold, the overall separation is clear. The violin plot further confirms that misclassifications are localized around the decision boundary.

To further analyze the model's explainability, we selected the known antiviral peptide LEDGF 361–370 (UniProt: O75475) (Hayouka et al., 2010) for case analysis. This peptide inhibits the catalytic activity of the HIV-1 integrase (IN) protein and effectively suppresses HIV-1 replication in cells. Notably, the tryptophan (W) residue in this sequence possesses unique fluorescent properties. In the study by Hayouka Z et al. (Hayouka et al., 2010), fluorescence anisotropy was employed to measure the binding affinity of peptide derivatives to IN following residue-by-residue substitutions. In our study, we utilized the sequence LEDGF 361–370 to validate the effectiveness of the model. As shown in Fig. 4C, the entire magenta-colored region corresponds to residues 361–370. The sequence (NSLKIDNLDV) had a predicted probability of 0.9562, which is far above the threshold, and the prediction was classified as a positive sample. Furthermore, we analyzed the model's attention mechanism to investigate its focus on different positions within the sequence (Fig. 4D). The results show that the model pays higher attention to positions 363, 364, and 366, which aligns with findings from previous studies. These studies indicated that when the amino acids at these positions were substituted, the derivatives exhibited significantly reduced inhibition of IN and a marked impact on AVP affinity. This suggests that the model, through ESM-2, has successfully learned the critical structural features of amino acids within the sequence. In contrast, the model showed lower attention to positions 362 and 370, consistent with findings that substitutions at these positions resulted in higher Kd inhibition concentrations and less reduction in affinity. Additionally, the residues at positions 365 and 366, which are critical binding sites for protein interactions, were also highlighted by the model. These results suggest that the attention distribution is partially consistent with experimentally reported functional residues, providing supportive evidence that FusionAVP may capture biologically

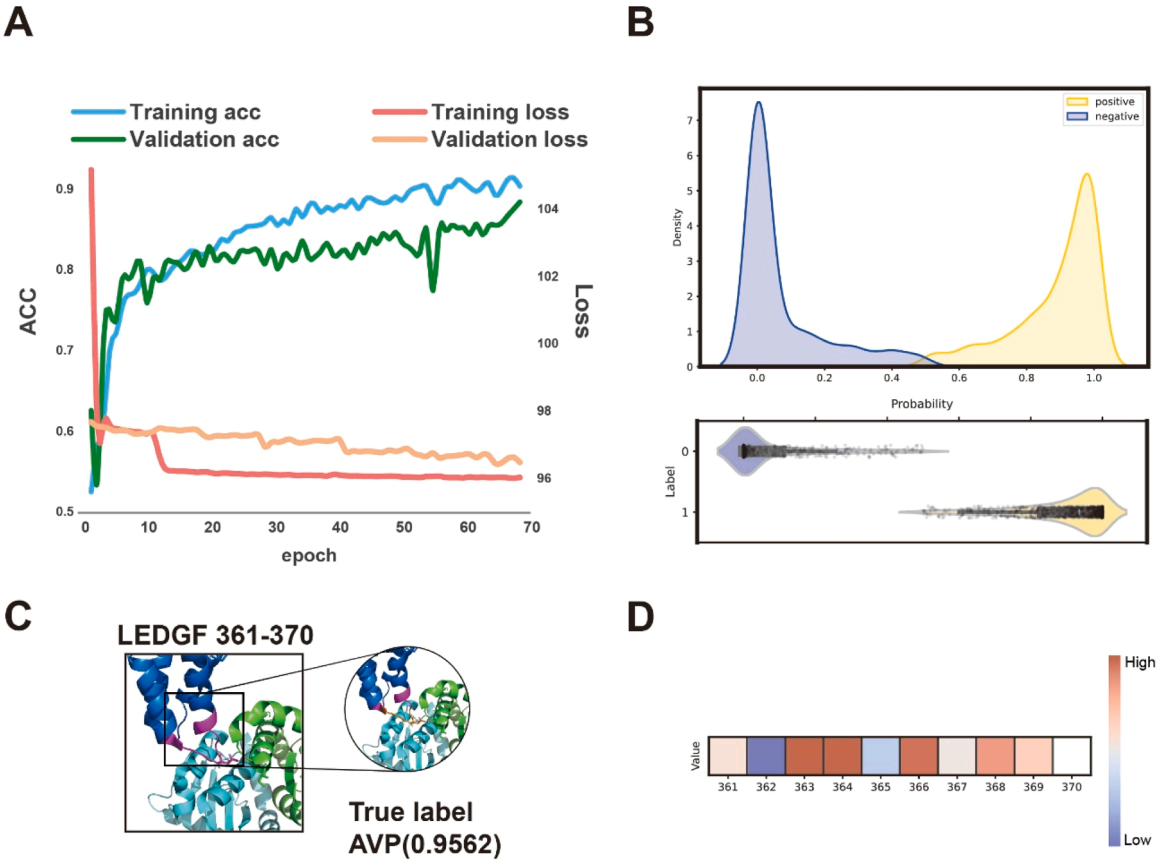


Fig. 4. Interpretability analysis through model performance and case studies. **A.** We visualize the model training and validation process, demonstrating the convergence of the loss. **B.** A kernel density plot, followed by the presentation of the predicted probabilities on the independent test set, displayed through a violin plot to illustrate the specific effects. **C.** Case study analysis using HIV antiviral peptides, where our model predicts a positive sample with a probability of 0.9562 (greater than 0.5, indicating an AVP). **D.** Visualization of the attention mechanism, highlighting the sites the model focuses on.

relevant positions.

To assess the sensitivity of cross-attention score profiles to sequence perturbations, we introduced random amino acid substitutions into the antiviral peptide NSLKIDNLDV. Figure S1 shows the original sequence and two perturbed examples, with substitutions at position 365 and at positions 363 and 368, respectively. Following substitution at position 365, its relative score increased, whereas the prominent scores at positions 364 and 366 in the original sequence became less pronounced. Following substitutions at positions 363 and 368, the relative scores at both positions decreased, accompanied by changes at other, unmodified positions. These observations indicate that the score profile of the final cross-attention layer responds to changes in the input sequence, with effects extending beyond the substituted positions, reflecting its sensitivity to sequence context.

3.6. Web server

A publicly accessible and user-friendly web server can greatly facilitate the practical application of bioinformatics tools by lowering technical barriers. To this end, we developed an intuitive web server for deploying our proposed model, FusionAVP, available at: http://www.bioai-lab.com/FusionAVP. As shown in Fig. 1F, users can submit sequences in FASTA format either by entering them directly into the text box or by uploading a FASTA file. Each sequence must contain no >100 amino acid residues. For format guidance, the "Example Sequence" button provides a valid sample. Upon clicking "Submit," the server returns both the predicted classification and the corresponding confidence scores.

4. Conclusion

In this study, we proposed a novel model, FusionAVP, based on a cross-attention-driven heterogeneous features fusion strategy, designed for the identification and prediction of antiviral peptides. The primary goal of this model is to capture the relationship between the independent view of traditional features and the encoded information extracted by large-scale models. By leveraging the proposed strategy, FusionAVP effectively integrates the feature representations of both views, thereby enhancing the classification performance of AVPs and offering insights for the classification and prediction of other biological sequences. Experimental results on publicly available datasets demonstrate that FusionAVP outperforms existing models. Additionally, ablation studies confirm the effectiveness of the proposed feature fusion module. In the case studies, FusionAVP successfully predicted unknown antiviral peptides, showcasing its strong efficacy and robustness. Furthermore, the attention mechanism matrices provided biologically meaningful explanations, shedding light on the underlying biological significance of the model's decisions.

Despite the excellent performance of our model, there remains room for further improvement. In future work, we will further collect AVPs with experimentally validated binding sites, residue-substitution data, or structural information to systematically evaluate the attention-based interpretation results. To comprehensively validate the effectiveness of the feature fusion module, it would be beneficial to extend the approach to other types of bioactive peptides. Currently, our work primarily focuses on using a single category of traditional features for classification, without fully exploring the complementarity among different traditional features. In future work, we plan to further explore the complementarity among traditional features by leveraging the proposed feature fusion module. Another limitation of this study is that the training and test datasets were constructed in a balanced manner to facilitate fair model evaluation, whereas real-world AVP screening scenarios are often highly imbalanced, with true AVPs representing only a small proportion of candidate peptides. Such imbalance may influence the sensitivity-specificity trade-off and the optimal decision threshold in practical applications. In addition, although the negative samples used in this study were filtered to reduce sequence redundancy, the negative sampling strategy may still introduce potential bias, because sampled non-AVPs may not fully represent the diversity of the real non-AVP peptide space. Future work will evaluate FusionAVP under more realistic imbalanced settings and explore more diverse, biologically representative negative sample construction strategies to improve model generalizability. Additionally, we aim to incorporate AlphaFold3 to extract 3D structural information and utilize the model to enhance the generalizability for identifying and classifying a broader range of bioactive peptides.

Funding

The work is supported by the National Natural Science Foundation of China (No. 62450002), the Science and Technology special fund of Hainan Province (ZDYF2024GXJS018), the Qiqihar Medical Institute Foundation (QMSI2024Z-01), the Education Department Foundation of Heilongjiang Province (2024-KYYWF-0339), the Scientific Technology Project of Qiqihar City (LSFGG-2024099) and the Graduate Research and Innovation Projects of Hainan Province (Qhys2024-231).

Data and software availability

The code is freely available at GitHub: https://github.com/aythunder/FusionAVP.

The users can access the server from http://www.bioai-lab.com/FusionAVP.

Ethics statement

The care and use of all individuals in the experiment comply with all Institutional and National Guidelines.

CRediT authorship contribution statement

Aoyun Geng: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization, Funding acquisition. **Jinwei Wang:** Software. **Chunyan Cui:** Investigation, Data curation. **Yuanhao Li:** Investigation, Data curation. **Junlin Xu:** Writing – review & editing, Supervision. **Yajie Meng:** Writing – review & editing, Supervision. **Leyi Wei:** Writing – review & editing, Supervision. **Quan Zou:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Zilong Zhang:** Writing – review & editing, Supervision, Conceptualization. **Tao Wang:** Writing – review & editing, Supervision, Funding acquisition. **Feifei Cui:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neunet.2026.109845.

References

Aguilera-Mendoza, L., Ayala-Ruano, S., Martinez-Rios, F., Chavez, E., García-Jacas, C. R., Brizuela, C. A., & Marrero-Ponce, Y. (2023). StarPep toolbox: An open-source software to assist chemical space analysis of bioactive peptides and their functions using complex networks. *Bioinformatics, 39*(8), Article bTad506.

Akbar, S., Ali, F., Hayat, M., Ahmad, A., Khan, S., & Gul, S. (2022). Prediction of antiviral peptides using transform evolutionary & SHAP analysis based descriptors by

incorporation with ensemble learning strategy. *Chemometrics and Intelligent Laboratory Systems, 230*, Article 104682.
Akbar, S., Raza, A., & Zou, Q. (2024). Deepstacked-AVPs: Predicting antiviral peptides using tri-segment evolutionary profile and word embedding based multi-perspective features with deep stacking model. *BMC Bioinformatics [Electronic Resource], 25*(1), 102.
Breitbart, M., & Rohwer, F. (2005). Here a virus, there a virus, everywhere the same virus? *Trends in Microbiology, 13*(6), 278–284.
Cao, R., Hu, W., Wei, P., Ding, Y., Bin, Y., & Zheng, C. (2023). FFMAVP: A new classifier based on feature fusion and multitask learning for identifying antiviral peptides and their subclasses. *Briefings in Bioinformatics, 24*(6), Article bBad353.
Chang, K. Y., & Yang, J.-R. (2013). Analysis and prediction of highly effective antiviral peptides based on random forests. *PLoS ONE, 8*(8), Article E70166.
Consortium, U. (2015). UniProt: A hub for protein information. *Nucleic Acids Research, 43*(D1), D204–D212.
Du, Z., Ding, X., Hsu, W., Munir, A., Xu, Y., & Li, Y. (2024). PLM4ACE: A protein language model based predictor for antihypertensive peptide screening. *Food Chemistry, 431*, Article 137162.
Du, Z., Ding, X., Xu, Y., & Li, Y. (2023). UniDL4BioPep: A universal deep learning architecture for binary classification in peptide bioactivity. *Briefings in Bioinformatics, 24*(3), Article bBad135.
Elnaggar, A., Heinzinger, M., Dallago, C., Rehawi, G., Wang, Y., Jones, L., Gibbs, T., Feher, T., Angerer, C., & Steinegger, M. (2021). Prottrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 44*(10), 7112–7127.
Fang, Y., Xu, F., Wei, L., Jiang, Y., Chen, J., Wei, L., & Wei, D.-Q. (2023). AFP-MFL: Accurate identification of antifungal peptides using multi-view feature learning. *Briefings in Bioinformatics, 24*(1), Article bBac606.
Fernandez-Diaz, R., Cossio-Pérez, R., Agoni, C., Lam, H. T., Lopez, V., & Shields, D. C. (2023). AutoPeptideML: Automated machine learning for building trustworthy peptide bioactivity predictors. *BioRxiv*, 2023.2011.2013.566825.
Fu, L., Niu, B., Zhu, Z., Wu, S., & Li, W. (2012). CD-HIT: Accelerated for clustering the next-generation sequencing data. *Bioinformatics, 28*(23), 3150–3152.
Fu, X., Yuan, Y., Qiu, H., Suo, H., Song, Y., Li, A., Zhang, Y., Xiao, C., Li, Y., Dou, L., et al. (2024). AGF-PPIS: A protein–protein interaction site predictor based on an attention mechanism and graph convolutional networks. *Methods, 222*, 142–151.
García-Jacas, C. R., García-González, L. A., Martinez-Rios, F., Tapia-Contreras, I. P., & Brizuela, C. A. (2022). Handcrafted versus non-handcrafted (self-supervised) features for the classification of antimicrobial peptides: Complementary or redundant? *Briefings in Bioinformatics, 23*(6), Article bBac428.
Geng, A., Luo, Z., Li, A., Zhang, Z., Zou, Q., Wei, L., & Cui, F. (2025). ACP-CLB: An anticancer peptide prediction model based on multichannel discriminative processing and integration of large pretrained protein language models. *Journal of Chemical Information and Modeling, 65*(5), 2336–2349.
Guan, J., Yao, L., Xie, P., Chung, C.-R., Huang, Y., Chiang, Y.-C., & Lee, T.-Y. (2024). A two-stage computational framework for identifying antiviral peptides and their functional types based on contrastive learning and multi-feature fusion strategy. *Briefings in Bioinformatics, 25*(3), Article bBae208.
Han, K., Wang, Y., Chen, H., Chen, X., Guo, J., Liu, Z., Tang, Y., Xiao, A., Xu, C., & Xu, Y. (2022). A survey on vision transformer. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 45*(1), 87–110.
Hayouka, Z., Levin, A., Maes, M., Hadas, E., Shalev, D. E., Volsky, D. J., Loyter, A., & Friedler, A. (2010). Mechanism of action of the HIV-1 integrase inhibitory peptide LEDGF 361–370. *Biochemical and Biophysical Research Communications, 394*(2), 260–265.
Henikoff, S., & Henikoff, J. G. (1992). Amino acid substitution matrices from protein blocks. *Proceedings of the National Academy of Sciences, 89*(22), 10915–10919.
Huang, Z., Wang, X., Huang, L., Huang, C., Wei, Y., & Liu, W. (2019). Ccnet: Criss-cross attention for semantic segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 603–612).
Javanian, M., Barary, M., Ghebrehewet, S., Koppolu, V., Vasigala, V., & Ebrahimpour, S. (2021). A brief review of influenza virus infection. *Journal of Medical Virology, 93*(8), 4638–4646.
Jia, Y., Zhang, Z., Yan, S., Zhang, Q., Wei, L., & Cui, F. (2024). Voting-ac4C: Pre-trained large RNA language model enhances RNA N4-acetylcytidine site prediction. *International Journal of Biological Macromolecules, 282*, Article 136940.
Kawashima, S., Pokarowski, P., Pokarowska, M., Kolinski, A., Katayama, T., & Kanehisa, M. (2007). AAindex: Amino acid index database, progress report 2008. *Nucleic Acids Research, 36*(suppl_1), D202–D205.
Kurata, H., Tsukiyama, S., & Manavalan, B. (2022). IACVP: Markedly enhanced identification of anti-coronavirus peptides using a dataset-specific word2vec model. *Briefings in Bioinformatics, 23*(4), Article bBac265.
Li, H., Wu, B., Sun, M., Zhu, Z., Chen, K., & Ge, H. (2024). Cross-domain contrastive graph neural network for lncRNA–protein interaction prediction. *Knowledge-Based Systems, 296*, Article 111901.
Li, Y., Wei, X., Yang, Q., Xiong, A., Li, X., Zou, Q., Cui, F., & Zhang, Z. (2024). MsBERT-promoter: A multi-scale ensemble predictor based on BERT pre-trained model for the two-stage prediction of DNA promoters and their strengths. *BMC Biology, 22*(1), 126.
Liang, Y., Ma, X., Li, J., & Zhang, S. (2025;). IACVP-MR: Accurate identification of anti-coronavirus peptide based on multiple features information and recurrent neural network. *Current Medicinal Chemistry, 32*(10), 2055–2067.
Lin, T.-T., Sun, Y.-Y., Wang, C.-T., Cheng, W.-C., Lu, I.-H., Lin, C.-Y., & Chen, S.-H. (2022). AI4AVP: An antiviral peptides predictor in deep learning approach with generative adversarial network data augmentation. *Bioinformatics Advances, 2*(1), Article vBac080.
Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., Smetanin, N., Verkuil, R., Kabeli, O., & Shmueli, Y. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science, 379*(6637), 1123–1130.
Lissabet, J. F. B., Belén, L. H., & Farias, J. G. (2019). AntiVPP 1.0: A portable tool for prediction of antiviral peptides. *Computers in Biology and Medicine, 107*, 127–130.
Luo, Y., Shi, L., Li, Y., Zhuang, A., Gong, Y., Liu, L., & Lin, C. (2025). From intention to implementation: Automating biomedical research via LLMs. *Science China Information Sciences, 68*(7), Article 170105.
McInnes, L., Healy, J., & Melville, J.:(2018). Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint* arXiv:180203426.
Nath, A. (2021). Prediction for understanding the effectiveness of antiviral peptides. *Computational Biology and Chemistry, 95*, Article 107588.
Pang, Y., Yao, L., Jhong, J.-H., Wang, Z., & Lee, T.-Y. (2021). AVPIden: A new scheme for identification and functional prediction of antiviral peptides based on machine learning approaches. *Briefings in Bioinformatics, 22*(6), Article bBab263.
Perez, R., Li, X., Giannakoulias, S., & Petersson, E. J. (2023). AggBERT: Best in class prediction of hexapeptide amyloidogenesis with a semi-supervised ProtBERT model. *Journal of Chemical Information and Modeling, 63*(18), 5727–5733.
Pinacho-Castellanos, S. A., García-Jacas, C. R., Gilson, M. K., & Brizuela, C. A. (2021). Alignment-free antimicrobial peptide predictors: Improving performance by a thorough analysis of the largest available data set. *Journal of Chemical Information and Modeling, 61*(6), 3141–3157.
Qureshi, A., Thakur, N., Tandon, H., & Kumar, M. (2014). AVPdb: A database of experimentally validated antiviral peptides targeting medically important viruses. *Nucleic Acids Research, 42*(D1), D1147–D1153.
Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., Guo, D., Ott, M., Zitnick, C. L., & Ma, J. (2021). Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences, 118*(15), Article e2016239118.
Schaduangrat, N., Nantasenamat, C., Prachayasittikul, V., & Shoombuatong, W. (2019). Meta-iAVP: A sequence-based meta-predictor for improving the prediction of antiviral peptides using effective feature representation. *International Journal of Molecular Sciences, 20*(22), 5743.
Sharma, R., Shrivastava, S., Singh, S. K., Kumar, A., Singh, A. K., & Saxena, S. (2021). Deep-AVPpred: Artificial intelligence driven discovery of peptide drugs for viral infections. *IEEE Journal of Biomedical and Health Informatics, 26*(10), 5067–5074.
Skalickova, S., Heger, Z., Krejcova, L., Pekarik, V., Bastl, K., Janda, J., Kostolansky, F., Vareckova, E., Zitka, O., & Adam, V. (2015). Perspective of use of antiviral peptides against influenza virus. *Viruses., 7*(10), 5428–5442.
Suzek, B. E., Huang, H., McGarvey, P., Mazumder, R., & Wu, C. H. (2007). UniRef: Comprehensive and non-redundant UniProt reference clusters. *Bioinformatics, 23*(10), 1282–1288.
Thakur, N., Qureshi, A., & Kumar, M. (2012). AVPpred: Collection and prediction of highly effective antiviral peptides. *Nucleic Acids Research, 40*(W1), W199–W204.
Tung, C.-H., Huang, J.-W., & Yang, J.-M. (2007). Kappa-alpha plot derived structural alphabet and BLOSUM-like substitution matrix for rapid search of protein structure database. *Genome Biology, 8*, 1–16.
Ullah, M., Akbar, S., Raza, A., & Zou, Q. (2024). DeepAVP-TPPred: Identification of antiviral peptides using transformed image-based localized descriptors and binary tree growth algorithm. *Bioinformatics, 40*(5), Article bTae305.
Vilas Boas, L. C. P., Campos, M. L., Berlanda, R. L. A., de Carvalho Neves, N., & Franco, O. L. (2019). Antiviral peptides as promising therapeutic drugs. *Cellular and Molecular Life Sciences, 76*, 3525–3542.
Wang, P., Zheng, S., Jiang, Y., Li, C., Liu, J., Wen, C., Patronov, A., Qian, D., Chen, H., & Yang, Y. (2022). Structure-aware multimodal deep learning for drug–protein interaction prediction. *Journal of Chemical Information and Modeling, 62*(5), 1308–1317.
Wang, X., Ding, Z., Wang, R., & Lin, X. (2023). Deepro-Glu: Combination of convolutional neural network and Bi-LSTM models using ProtBert and handcrafted features to identify lysine glutarylation sites. *Briefings in Bioinformatics, 24*(2), Article bBac631.
Wang, Y., Zhai, Y., Ding, Y., & Zou, Q. (2024). SBSM-Pro: Support bio-sequence machine for proteins. *Science China Information Sciences, 67*(11), Article 212106.
Wei, L., Zhou, C., Su, R., & Zou, Q. (2019). PEPred-Suite: Improved and robust prediction of therapeutic peptides using adaptive feature representation learning. *Bioinformatics, 35*(21), 4272–4280.
Wu, H., Liu, J., Jiang, T., Zou, Q., Qi, S., Cui, Z., Tiwari, P., & Ding, Y. (2024). AttentionMGT-DTA: A multi-modal drug-target affinity prediction using graph transformer and attention mechanism. *Neural Networks, 169*, 623–636.
Xiao, C., Zhou, Z., She, J., Yin, J., Cui, F., & Zhang, Z. (2024). PEL-PVP: Application of plant vacuolar protein discriminator based on PEFT ESM-2 and bilayer LSTM in an unbalanced dataset. *International Journal of Biological Macromolecules, 277*, Article 134317.
Xu, D., Chen, W., Peng, W., Zhang, C., Xu, T., Zhao, X., Wu, X., Zheng, Y., Wang, Y., & Chen, E. (2024). Large language models for generative information extraction: A survey. *Frontiers of Computer Science, 18*(6), Article 186357.
Yan, C., Geng, A., Pan, Z., Zhang, Z., & Cui, F. (2024). MultiFeatVotPIP: A voting-based ensemble learning framework for predicting proinflammatory peptides. *Briefings in Bioinformatics, 25*(6), Article bBae505.
Yin, J., Wu, H., & Sun, S. (2023). Effective sample pairs based contrastive learning for clustering. *Information Fusion, 99*, Article 101899.
Zhu, H., Yang, Y., Wang, Y., Wang, F., Huang, Y., Chang, Y., Wong, K.-C., & Li, X. (2023). Dynamic characterization and interpretation for protein-RNA interactions across diverse cellular conditions using HDRNet. *Nature Communications, 14*(1), 6824.